



M.A.M. SCHOOL OF ENGINEERING

(An Autonomous Institution)

Accredited by NAAC | Approved by AICTE-New Delhi | Affiliated to Anna University-Chennai
Trichy-Chennai Trunk Road, Siruganur, Tiruchirappalli -621 105



Department of Electrical and Electronics Engineering Academic

Year 2025-2026 (Even Semester)

Unit - I

Sub.Code/Sub.Name : EE3602 Power System Operation and Control Date :
Year/Sem. : III / VI Duration : 3 Hrs
Max.Marks : 100

	PART A	M.A	CO	BTL
1.	Define spinning reserve.	2	CO1	BT1
2.	What is diversity factor.	2	CO1	BT1
3.	Differentiate load curve and load duration curve.	2	CO1	BT2
4.	Show the conditions necessary for sharing load operating in parallel between the two synchronous machines?	2	CO1	BT1
5.	State area control error.	2	CO1	BT1
6.	Demonstrate the term average load.	2	CO1	BT1
7.	Mention different types of to find load forecasting	2	CO1	BT2
8.	What is the necessity to regulate voltage and frequency in the power system?	2	CO1	BT1
9.	Define load factor	2	CO1	BT1
10.	Quote about coherent group of generators	2	CO1	BT2

	Part B	M.A	CO	BTL												
11.	Draw the P - f and Q - V control channel of a synchronous generator and explain how voltage and frequency are maintained constant	16	CO1	BT2												
12.	What is the necessity of load forecasting and explain different methods load forecasting techniques in power system	16	CO1	BT2												
13.	Briefly discuss the various types of load curves and basics of electrical tariff	16	CO1	BT4												
14.	Describe about Plant Level control and System level control	16	CO1	BT4												
15.	A generating station has the following daily load cycle:<table><tr><td>Time (hours): 0-6</td><td>6-10</td><td>10-12</td><td>12-16</td><td>16-20</td><td>20-24</td></tr><tr><td>Load (MW): 20</td><td>25</td><td>30</td><td>25</td><td>35</td><td>20</td></tr></table> Construct the load curve and evaluate, 1. Maximum demand 2. Units generated/day 3. Average load 4. Load factor.	Time (hours): 0-6	6-10	10-12	12-16	16-20	20-24	Load (MW): 20	25	30	25	35	20	16	CO1	BT4
Time (hours): 0-6	6-10	10-12	12-16	16-20	20-24											
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Department of Electrical and Electronics Engineering

Academic Year 2025-2026 (Even Semester)

Unit - II

Sub.Code/Sub.Name : EE3602 Power System Operation and Control

Date :

Year/Sem. : III / VI

Duration : 3 Hrs

Max.Marks : 100

	PART A	M.A	CO	BTL
1.	What is the function of load frequency control?	2	CO2	BT2
2.	State any two necessities to put alternator in parallel?	2	CO2	BT1
3.	Define per unit droop or speed regulation?	2	CO2	BT2
4.	Write the tie-line power deviation equation in terms of frequency?	2	CO2	BT4
5.	What is meant by control area?	2	CO2	BT2
6.	Differentiate static and dynamic response an ALFC loop?	2	CO2	BT4
7.	What are the types of automatic load frequency control for interconnected power system?	2	CO2	BT2
8.	Explain the principles of tie-line bias control?	2	CO2	BT2
9.	What is area control error?	2	CO2	BT2
10.	What is the function of load frequency control?	2	CO2	BT2

	Part B	M.A	CO	BTL
11.	What are the components of speed governing mechanism and derive it's transfer function with an aid of a block diagram?	13	CO2	BT4
12.	Discuss about Single Area static analysis of controlled and uncontrolled case	13	CO2	BT4
13.	Explain the Tie-line with frequency bias control of Two Area system	13	CO2	BT4
14.	Derive the expression for state variable model of single area LFC system with neat block diagram	13	CO2	BT4
15.	Derive the modelling of two area load frequency control system with neat block diagram.	13	CO2	BT4



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Department of Electrical and Electronics Engineering Academic

Year 2025-2026 (Even Semester)

Unit - III

Sub.Code/Sub.Name : EE3602 Power System Operation and Control

Date :

Year/Sem. : III / VI

Duration : 3 Hrs

Max.Marks : 100

	PART A	M.A	CO	BTL
1.	Distinguish between on load and off load tap changing	2	CO3	BT1
2.	What are the static shunt compensator?	2	CO3	BT1
3.	What are the methods of voltage control?	2	CO3	BT2
4.	Compare shunt and series capacitor s	2	CO3	BT1
5.	What is SVC? Why is it used?	2	CO3	BT1
6.	Draw the phasor diagram for static VAR compensators.	2	CO3	BT1
7.	Discuss the effect of common series on the maximum power transfer in a transmission line	2	CO3	BT2
8.	Describe the role of synchronous condenser and static VAR system in voltage control	2	CO3	BT1
9.	What are the advantage and disadvantage of synchronous compensator	2	CO3	BT1
10	What is mean by regulating transformer	2	CO3	BT2

		Part B	M.A	CO	BTL
11.	a)	Explain the operations of brushless AC excitation system AVR loop system	16	CO3	BT2
12.	a)	Explain the Automatic Voltage Regulator and static and dynamic analysis	16	CO3	BT2
13.	a)	Classify the different methods of reactive power injections and compare them	16	CO3	BT4
14.	a)	Explain the SVC and STATCOM for voltage control	16	CO3	BT4
15.	a)	Compare the operation of an on load and off load tap changing transformer	16	CO3	BT4



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Year 2025-2026 (Even Semester)

Unit - IV

Sub.Code/Sub.Name : EE3602 Power System Operation and Control

Date :

Year/Sem. : III / VI

Duration : 3 Hrs

Max.Marks : 100

	PART A	M.A	CO	BTL
1.	What is the purpose of economic dispatch?	2	CO3	BT1
2.	What are the factor of affecting the cost of generation?	2	CO3	BT1
3.	What is Lagrangian multiplier?	2	CO3	BT2
4.	Explain the terms incremental operating cost of a power system	2	CO3	BT1
5.	Write the coordination equations of neglecting losses	2	CO3	BT1
6.	Draw incremental fuel cost curve for a thermal plant	2	CO3	BT1
7.	Compare with unit commitment and economical load dispatch	2	CO3	BT2
8.	What are the losses of coefficient?	2	CO3	BT1
9.	Define incremental efficiency	2	CO3	BT1
10.	What is the participation factor with respect to economical load dispatch?	2	CO3	BT2

	Part B	M.A	CO	BTL
11.	Explain Lambda iteration algorithm for solving economic scheduling problem without transmission loss.	16	CO3	BT2
12.	Examine the coordination equation for economic dispatch neglecting losses.	16	CO3	BT2
13.	The fuel cost functions for three thermal plants in \$/h are given by $F_1 = 0.0045P_{g1}^2 + 5.2P_{g1} + 580$ $F_2 = 0.0056P_{g2}^2 + 4.5P_{g2} + 640$ $F_3 = 0.0079P_{g3}^2 + 5.8P_{g3} + 820$ where, P_{g1}, P_{g2}, P_{g3} are in MW. Find out the optimal dispatch and the total cost when the total load is 925 MW with the following generator limits. $250MW \leq P_{g1} \leq 450MW$, $200MW \leq P_{g2} \leq 350MW$, $125MW \leq P_{g3} \leq 250MW$	16	CO3	BT4



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14.	Explain the mathematical formulation of long-term hydro-thermal scheduling.	16	CO3	BT4
15.	The input output curve characteristics of three units are $F_1=750+6.49P_{g1}+0.0035P_{g1}^2$ $F_2=870+5.75P_{g2}+0.0015P_{g2}^2$ $F_3=620+8.56P_{g3}+0.001P_{g3}^2$ The fuel cost of unit 1 is 1.0 Rs/MBtu, 1.0 s/MBtu for unit 2 and 1.0 s/MBtu for unit 3. Total load is 800MW. Use the participation factor method to Calculate the dispatch for a load is increased to 880MW?	16	CO3	BT4



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Year 2025-2026 (Even Semester)

Unit - V

Sub.Code/Sub.Name : EE3602 Power System Operation and Control

Date :

Year/Sem. : III / VI

Duration : 3 Hrs

Max.Marks : 100

	PART A	M.A	CO	BTL
1.	What is the need of computer control of power system?	2	C03	BT1
2.	What is EMS and major functions of EMS,	2	C03	BT1
3.	Define emergency control centre	2	C03	BT2
4.	What are the functions of SCADA?	2	C03	BT1
5.	What are the different operating States power system can operate in?	2	C03	BT1
6.	List the factor that effects the power system security	2	C03	BT1
7.	What are the takes of energy control centre	2	C03	BT2
8.	Define state estimation	2	C03	BT1
9.	What is load forecasting?	2	C03	BT1
10.	Compare control strategies action by operator and variables to be adjusted	2	C03	BT2

		Part B	M.A	CO	BTL
11.	a)	Explain the hard work components and functional aspects of SCADA system using a fundamental block diagram	16	C03	BT2
12.	a)	Explain the deterministic state Estimation weighted least square Estimation .	16	C03	BT2
13.	a)	Explain the various functions of Energy control centre	16	C03	BT4
14.	a)	Analyze the need for computer control of power system.	16	C03	BT4
15.	a)	Explain the Measurements and Errors and Virtual Measurements	16	C03	BT4
16.	a)	Explain in detail about state estimation.	16	C03	BT4